+Colabor Creative Coding

Process Documentation, Lukas Bucher

Sprint 1: Re-Code Re-Think

For our first sprint, we chose to re-code MMVII. Gesundheit, Glück und Geld im Neuen Jahr (2006) by Anton Bruhin. We were immediately drawn to the visuals of the work, especially its abstract compositions and pixelated textures. The piece is eye-catching and creates a strong foundation to build on. The fact that it includes several components and opportunities to reimagine the work was one of the main reasons why we chose it. Each of us could select an aspect of the work and transform it into something new through code while experimenting based on personal interests and skill levels.

We could not find a lot of information about this specific work in our research, so we can't say for sure how exactly Bruhin created it. We found that he had created similar pieces using Microsoft Paint or Microsoft Word, and we assume this one was made with similar digital tools.

What fascinated me most was the visual effect that resembles the interference seen when photographing a digital screen. Through research I found that this phenomenon is known as the moiré effect and it occurs when two similar patterns overlap at slight offsets, creating new, often surprising patterns. Bruhin's work captures this sensation within a static pixel grid, simulating motion and depth through simple repetition or inversion.

I started by building a pixel grid, since the work clearly relies on pixel-based patterns. My first step was recreating a background of black and white horizontal stripes. I then experimented with two main techniques: obstruction, which turns areas into vertical black and white stripes based on adjacent pixel colors, and inversion, which flips pixel colors.

At first, I applied these to rectangles, but since many shapes in Bruhin's work appear curved, I extended the approach to circles. This was more challenging due to pixel calculations. By combining obstruction and inversion, I could create an illusion of depth, making circles appear like raised buttons.

Next, I experimented with animation, repetition, and background variations. One of the most visually exciting effects came from repeatedly inverting slightly offset circles. This layering technique created visuals similar to the moiré patterns that initially caught my attention.

Finally, I explored interactivity by adding a real-time camera layer. In this version, the darker areas of the live feed are dynamically obstructed, resulting in a responsive, evolving graphic.

Sprint 2: Take a Break

In Sprint 2, we were tasked with creating a browser extension around the theme *Take a Break*. We worked in a team, but since our coding skill levels varied, we started by experimenting individually. While others were being introduced to HTML, CSS, and JavaScript, I began developing the first extension, initially targeting Firefox and later adapting it for Chrome.

My first idea was to transform websites into a guided breathing exercise, encouraging users to pause and breathe. I used CSS animations and transitions to manipulate any webpage. The site zooms in and out, elements expand and shrink, opacity and letter spacing pulse in rhythm, and text turns into the instructions *BREATHE IN* and *BREATHE OUT*. A blur appears around the screen edges to amplify the calming effect. I was happy with how this extension turned out, both visually and conceptually.

Since we were encouraged to try multiple ideas, I moved on to another concept the next day: breaking the website, giving the word *break* a double meaning. After breaking the site a certain amount, an interruption appears suggesting a break. We initially discussed having each teammate design their own unique interruption to combine later. I played with corrupting on-page text, making elements fall when clicked, and adding broken screen PNGs at click points to simulate damage.

I wanted to explore physics-based interaction, so I integrated Matter.js to simulate gravity and collision. After some trial and error, I got it working. DOM elements were tied to Matter.js physics bodies, letting them fall and interact naturally. This was both technically challenging and fun to play with.

Lastly, I wanted to learn how background scripts work. I built a usage-tracking extension that collects mouse movements and clicks across websites, visualizing them as a heatmap. Users can switch between data types and domains to gain insights into their browsing behavior.

Eventually, our team decided not to merge our extensions due to visual and conceptual differences. We agreed it would be better to present them individually and keep the option open to refine the ideas in the final sprint.

What worked well for me was trying many different directions. I learned a lot and enjoyed the process. However, a clearer team concept and more structure might have helped clarify where to put our focus towards the end of the sprint.

Sprint 3: Quick Response

In *Quick Response*, I explored the mobile phone as a personal, relatable, and responsive medium for interactive web-based experiments. The project consists of 16 small black-and-white pixel-style interactions, each accessible via a unique QR code. These QR codes are displayed together at the festival, forming a larger visual system that reflects the modular nature of digital experiences.

The title references both the QR code (Quick Response code) and the expectation of instant interaction that defines our relationship with digital media today. Each experiment plays with this expectation in different ways. Some respond immediately to user input, while others intentionally delay feedback or resist interaction altogether.

The experiments explore a wide range of input and sensor-based interactions, such as microphone volume, touch, motion, camera, multitouch, and waiting. For example, a moiré circle grows and shrinks based on ambient volume, a physics simulation responds to screen tilting, and a 10-minute timer challenges the user's patience with minimal feedback. Visually, the project builds on pixel aesthetics from my previous work, referencing both QR code structures and early digital graphics.

The work invites users to reflect on how quickly they expect the world to respond, both digitally and emotionally. It contrasts instant gratification with slow or ambiguous responses, raising questions about attention, habit, control, and the tension between action and reaction.

Festival Text

Quick Response is a series of 16 playful, browser-based mini experiments for your smartphone, each one accessed by scanning a different QR code.

The interactions are short, strange, and sometimes surprising. Some respond immediately to touch, movement, or sound, others require more patience. All are designed in a black-and-white pixel style, inspired by my take on Anton Bruhin's work and QR code aesthetics.

The project explores how we interact with our phones and how quickly we expect reactions from digital tools. It invites visitors to reflect on attention, responsiveness, and how even simple interactions can feel meaningful or absurd.

Technically, the pieces creatively use your phone's functions and sensors, like the microphone, camera, and accelerometer.

The idea grew from earlier experiments with pixel visuals and interactive browser extensions. Instead of focusing on one polished app, I chose to build many small sketches, each one testing a concept, interaction, or feeling.

Quick Response is playful, personal, and a bit unpredictable, just like the phone in your pocket.

Sprint 3 Pitch

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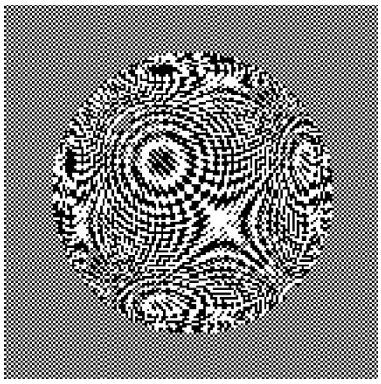
For Sprint 3, I want to focus on the **mobile phone** as the main platform. I find the device interesting because of its personal nature. It's something everyone carries with them and interacts with intimately. Since the final presentation will be at a festival, I think involving the visitors' own phones can create a stronger and more relatable experience.

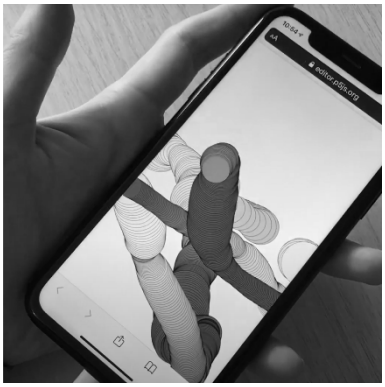
While I haven't fully decided on a specific direction yet, here are some ideas and themes I'd like to explore:

- **Sensors**: Use motion sensors (gyroscope, accelerometer) or device orientation to influence visuals or create reactive elements.
- **Camera**: Use the phone camera to overlay visual filters (e.g., pixelation, color shifting, moiré distortions) onto the real world.
- **Visuals from Sprint 1**: Reuse and reimagine pixel-based textures, grid structures, and moiré patterns from Sprint 1.
- **Physics from Sprint 2**: Implement physics interactions (e.g., falling elements, gravity) driven by real sensor data like phone tilting or shaking.
- **QR Codes**: Use a QR code as the entry point to the experience. I also want to experiment with the QR code as a visual element, inspired by the black-and-white pixel aesthetics of my earlier work. Maybe I could even generate visuals composed of QR codes (QR-codeception).

Inspiration



my work from Sprint 1



using motion sensors on mobile with p5.js

QR code art

Kurzdoku +Colabor Creative Coding, 06.06.2025

Lukas Bucher, Digital Ideation

Quick Response

Persönliches Interesse

In meinem finalen Projekt des Colabor-Moduls *Creative Coding* wollte ich herausfinden, wie das Smartphone für interaktive, visuelle Erlebnisse genutzt werden kann.

Ein Schwerpunkt lag auf den sensorischen Möglichkeiten des Handys (Kamera, Mikrofon, Bewegungssensoren) sowie QR-Codes als Schnittstelle zwischen physischer und digitaler Welt.

Arbeitsprozess

Der Arbeitsprozess gliederte sich in drei Sprints mit unterschiedlichen Schwerpunkten. Im ersten Sprint re-kodierte ich ein Werk von Anton Bruhin und entwickelte daraus ein interaktives Experiment in p5.js.

Im zweiten Sprint setzte ich mich mit Browsererweiterungen zum Thema *Take a Break* auseinander. Ich programmierte Extensions, die zu bewussten Pausen anregen.

Im dritten Sprint entstand das Hauptprojekt Quick Response: 16 kleine Webexperimente, aufrufbar über QR-Codes. Die Seiten nutzen Handysensoren wie Mikrofon, Kamera oder Bewegung und greifen gestalterisch auf die Pixeloptik des ersten Sprints sowie konzeptuelle Ideen aus dem zweiten zurück.

Fazit

Das Modul war eine gute Gelegenheit, technische Interessen mit künstlerischen Fragestellungen zu verbinden. Besonders lehrreich war es, mit Einschränkungen wie der limitierten Funktionalität von Mobilgeräten zu gestalten. Die interaktive Ausstellungssituation hat mir ausserdem geholfen, stärker aus Besuchersicht zu denken.